

Stray Grain Formation in the Seed Region of Single-Crystal Turbine Blades

X.L. Yang, P.D. Lee, and N. D'Souza

Seed crystals are frequently used to provide an off-axis $\langle 001 \rangle$ crystallographic orientation to investment cast single-crystal, nickel-based superalloy turbine blades. However, stray grain defects can form during the melt-back of the seed crystal, requiring the use of a helical grain selector between the seed and the blade to remove them. Using meso-scale numerical simulations, the

formation mechanisms of these stray grain defects have been investigated. Also investigated was the influence of the seed's crystallographic orientation relative to blade axis. The model is first validated by comparison to experimental observations and then by its application to a range of casting situations. The results show that initiation of these defects is difficult to avoid. Instead, the

impact of stray grains should be controlled during their growth.

INTRODUCTION

Modern turbine engine designs require the use of single-crystal (SX) superalloy turbine blades to obtain performance and life-cycle objectives.¹ Turbine blades in SX form are routinely cast from nickel-based superalloys. Single-crystal alloys offer improved creep-rupture, fatigue, oxidation, and coating properties, resulting in superior turbine engine performance and durability.^{2,3} However, the benefits of SX over conventionally cast and directionally solidified (DS) components critically depend on avoiding the introduction of casting defects, such as stray grains, freckles, or deviations from the required crystal orientation.⁴

Avoiding these defects can be difficult in blades where the service load is at an angle relative to the axis of the blade and hence the $\langle 001 \rangle$ crystallographic orientation of the blade is cast in alignment to the load. To do this, an SX seed of the desired off-axis orientation is used. The pre-fabricated seed is partially melted back during the soak period of the mold and subsequently molten alloy is introduced through the mold cavity across the seed melt-back interface. The mold is then withdrawn from the furnace.

During the initial stage of orientation transfer between the semi-solid seed and the molten alloy, nucleation of highly misoriented stray grains with respect to the SX axis have been observed (see Figure 1a).⁵ Once nucleated, these stray grains may have a more favorable orientation for growth than the deliberately off-axis seed, potentially leading to their growth instead of the desired orientation, as shown in Figure 1b where many stray grains are found just above the melt-back interface (cross section i) with just a few

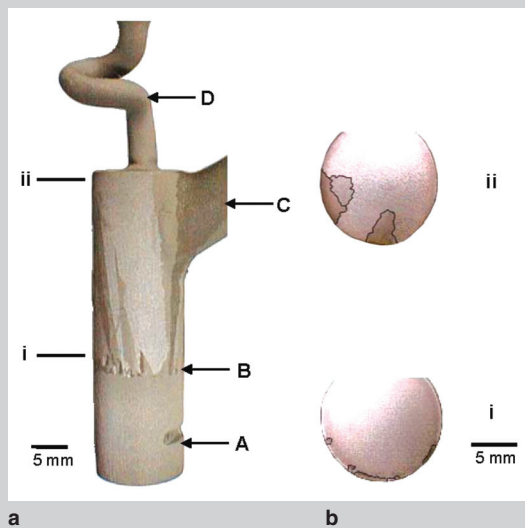


Figure 1. (a) The seed and helical selector section of a turbine blade casting macro-etched to highlight the stray grains; (b) optical micrographs of transverse sections showing the nucleation and growth of stray grains at 2.5 mm and 29.5 mm above the melt-back interface. A—seed; B—melt-back; C—runner feed; and D—helical selector leading to the turbine blade casting. Note: grain boundaries are highlighted in (b).⁵

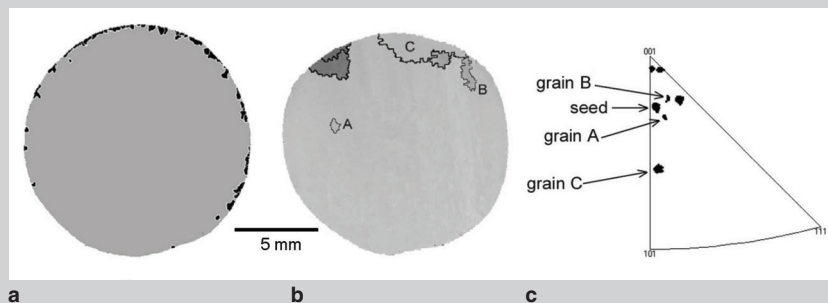


Figure 2. (a) An EBSD analysis of the transverse section ~2 mm ahead of the melt-back interface; (b) EBSD analysis of the transverse section 30 mm ahead of the melt-back interface illustrating only a few of the stray grains have survived after competitive growth; (c) inverse pole figure of the grain orientations from (b) relative to the casting direction.⁵

of the preferentially oriented stray grains growing at the expense of the seed grain just before entering the selector (cross section ii). The propensity for these stray grains to survive increases with increasing crystallographic misorientation of the seed. It is only the use of the helical selector that prevents these stray grains from entering the blade cavity. Although the helical selector is required to inhibit stray grains formed near the seed from growing into the component, inherent drawbacks are associated with its use, among them:

- Depending on the pitch of the spiral and its orientation with respect to the central sprue on the cluster mold, locally undercooled regions can occur resulting in new stray grains being nucleated.
- Cost is increased for several reasons: it increases the overall length of the assembly requiring longer DS furnaces; the selector can not be injection molded in wax at the same time as the blade, requiring expensive hand alignment of the seed together with the wax spiral and blade molds; and the investment mold is most susceptible to cracking in the selector region.

To eliminate the need for the spiral selector, it is crucial to understand the nucleation and growth of the stray grains

formed during seed melt-back so that their occurrence can be suppressed.

The nucleation mechanism of stray grains during the casting of metals had been shown to originate via two main mechanisms: heterogeneous nucleation and dendrite fragmentation.^{6,7} Stanford et al.⁵ tried to quantify the nucleation and growth of stray grains in the region of the seed experimentally by sequential sectioning of this region of the casting followed by electron back-scattered diffraction (EBSD) measurement of grain size and orientation. They observed that the nucleation of equiaxed stray grains occurred profusely around the perimeter of the seed within a distance of ≈ 2 mm from the melt-back interface (Figure 2a). This suggests that the nucleation of these stray grains is associated with the transient thermal fluctuations during melt-back/start-up. However, successive transverse sections moving away from the melt-back revealed that only a few of these grains survived, indicating that growth mechanisms also play a significant role (Figure 2b and 2c).

The objective of the study is to understand the possible mechanisms of stray grain nucleation and growth occurring during seed melt-back using meso-scale computational simulations. In particular, the roles of both the initial thermal transient and seed orientation are investigated using an in-house cel-

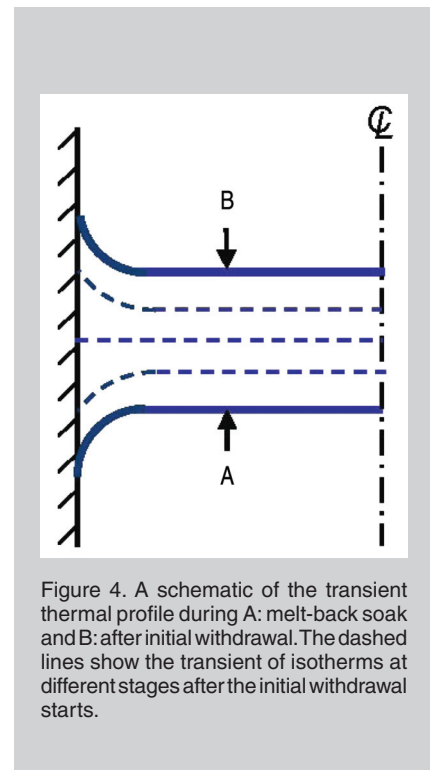


Figure 4. A schematic of the transient thermal profile during A: melt-back soak and B: after initial withdrawal. The dashed lines show the transient of isotherms at different stages after the initial withdrawal starts.

lular automaton-finite different model. This model was originally developed by Lee and co-workers to predict porosity in shape-cast aluminum alloys.^{8,9} This model has been extended to predict solidification defects in direct-chill cast wrought aluminum alloys¹⁰ and SX superalloys,¹¹ tree rings and grain evolution during vacuum-arc remelting,^{12,13} solidification microstructure,^{14,15} stray grain formation,¹⁶ and columnar to equiaxed transition in nickel-based superalloys and aluminum alloys.^{17,18}

The model was also extended into a multi-scale model by coupling with two different commercial finite-element software packages: ProCAST (ProCAST is registered by ESI Group, Paris, France), to predict solidification microstructure under complex casting geometry or thermal conditions where accurate temperature calculation was crucial,^{19,20} and ABAQUS (ABAQUS is registered by ABAQUS, Pawtucket, Rhode Island), to predict fatigue life of cast components as an explicit function of the microstructure to residual stress.²¹ In this study, the model was further extended to investigate the microstructure morphology and the formation of defect grains with varied seed orientation at the transient thermal stage during seed melt-back. Possible mechanisms for stray grain formation are examined and

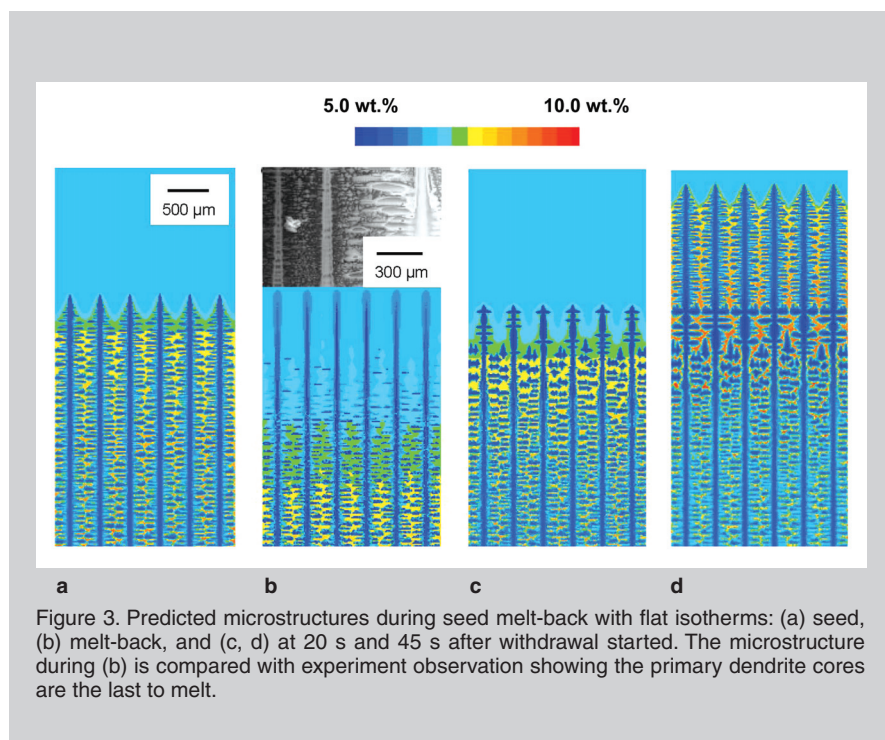


Figure 3. Predicted microstructures during seed melt-back with flat isotherms: (a) seed, (b) melt-back, and (c, d) at 20 s and 45 s after withdrawal started. The microstructure during (b) is compared with experiment observation showing the primary dendrite cores are the last to melt.

discussed. See the sidebar for a model description and simulation parameters.

SIMULATION OF SEED MELT-BACK

On partially remelting SX seeds of CMSX-4, islands of random orientation were observed within the semi-solid mush.³¹ Two main mechanisms have been suggested for the formation of these seeds: the melt back and pinch-off of secondary/tertiary arms and their subsequent rotation altering the crystallographic orientations, and heterogeneous nucleation in front or between the dendrite tips due to local undercooling. Although the latter mechanism has been investigated previously by many authors during advancing growth conditions,^{18,32–34} the thermal and solutal field produced are very different during seed melt-back. To determine the relative importance of the mechanisms, secondary/tertiary arm fragmentation is first examined alone by assuming that no heterogeneous nucleation occurs, followed by combined fragmentation and heterogeneous nucleation. The influence of isotherm shape is also studied separately by first implementing the flat isotherms that are typical of the center or for small-diameter seeds (<6 mm).³¹ The curved isotherms found at the mold wall are then implemented. Finally, the influence of off-axis <001> seed crystallographic orientations is explored. All of these parameters are discussed with respect to how they prevent or enhance the formation of stray grain defects.

Seed Melt-Back with Flat Isotherms

Before the melt-back can be simulated, an initial microstructure must be produced that is typical of the seed crystal. The seed is cut from the middle portion of an SX cylinder and hence will have a columnar dendritic structure. To replicate this in the model, grain nuclei were placed with the experimentally observed spacing at the bottom of the domain. Their growth was simulated for 40 s at a constant pulling velocity, reaching a steady state. Allowing this structure to act as the seed crystal (Figure 3a), the pulling velocity was then decelerated at a constant rate. It reached zero growth 5 s later.

Melt-back was then simulated by moving the isotherms downward by

2.25 mm (similar to industrial practice), raising the temperature around the tips of the primaries above their liquidus. The seed was then held in this temperature field for 10 s. As shown in Figure 3b, the upper portion of the primaries starts to melt. The secondaries just behind the primary tips melt due to their sharp curvature. Further down the primaries, the sharp curvature where the secondaries are attached to the primaries melts before the secondaries themselves, pinching them off from the primaries. These fragmented secondaries could easily change their crystallographic orientations due to local fluid flow. The cores of primaries are the last to melt due to their lower solute content, which is in good agreement with experiment observation (Figure 3b).³¹

After the soak period, the mold withdraw begins again (100 $\mu\text{m/s}$) and both the primaries and pinched-off second-

aries ripen and grow (Figure 3c). The predicted structure is very similar to those observed experimentally by Stanford et al.³¹ in samples quenched during melt-back, where a small amount of growth and coarsening occurred.

After a further 25 s growth, the pinched-off secondaries are overgrown by the original primaries (Figure 3d). They are far enough behind the primary tips, and in the solute-rich interdendritic regions, therefore their growth is slower. If fluid flow was incorporated into the simulations, these fragmented secondaries might alter their orientations, producing a number of small, randomly oriented grains as observed experimentally near the edge of the seed at the melt-back interface (Figure 2a). Although there are many new grains, after a relatively short distance the crystallographic orientation of the seed has been completely preserved.

MODEL DESCRIPTION AND SIMULATION PARAMETERS

A meso-scale cellular automaton-finite difference model was used to simulate the dendritic growth and heterogeneous nucleation during seed melt-back. The model solves for the solute diffusion-controlled growth of cubic structured alloys, predicting the growth of the columnar dendrites from the seed and their competitive growth interactions with stochastically nucleated stray grains. The stochastic nucleation is based on the local undercooling of the melt relative to the local liquidus temperature accounting for any solute enrichment at that location. The theory, implementation, and validation of this model are detailed in prior publications.^{9,14,17,19} Only the two features (melting and a more accurate surface curvature calculation) implemented to simulate seed melt-back are described, as well as the simulation parameters used in this study.

Melt-Back

As a reverse process of solidification, fraction solid change in an interface cell (implemented as a continuous function ranging from 0 to 1) is calculated in a similar way to that of a solidifying cell. During one time step, the variation of fraction solid can be deduced from^{8,22}

$$C_L(1-k) \frac{\partial f_s}{\partial t} = -\nabla \cdot (D_E \nabla C_L) + [1 - (1-k)f_s] \frac{\partial C_L}{\partial t} \quad (\text{A})$$

where C_L is the solute average concentration in the liquid, k the partition coefficient, f_s the fraction solid, and D_E the equivalent diffusion coefficient calculated as described elsewhere.^{14,22}

Equation A is used to obtain the fraction solid change, df_s , in the melting cells, and is solved using an explicit finite-difference scheme.

For a fully solid cell, a rule has to be defined to switch its status from solid into mushy. At each time step, the local temperature, T_c , in a solid cell (f_s equals to 1) is checked and its corresponding solidus temperature, T_s , is calculated from its solute concentration C_E . A small amount of superheating, ΔT , is considered to be necessary to overcome the energy barrier associated with melting. Therefore, the rule for a solid cell to change into a mushy cell is $T_c > T_s + \Delta T$. Once this condition is satisfied, the solute concentration in the liquid of the cell, C_L , can be derived from a linearized phase diagram and its local temperature

$$C_L = (T_c - T_L) / m + C_0 \quad (\text{B})$$

where C_0 is initial solute concentration, T_L liquidus temperature with concentration of C_0 , and m slope of liquidus from linearized phase diagram.

The fraction solid of the melting cell is calculated assuming solute conservation

$$f_s = (C_L - C_E) / (C_L^* - (1-k)) \quad (\text{C})$$

Seed Melt-Back with Curved Isotherms and Heterogeneous Nucleation

Simulations were carried out of the melt-back process near the surface of the mold wall including heterogeneous nucleation. During the soak period the mold is heated by the furnace, resulting in a higher temperature near the mold wall than in the center of the casting, producing convex isotherms near the wall (A in Figure 4). During initial withdrawal, the radial heat loss from the mold wall is greater than the center, producing concave isotherms near the wall (B in Figure 4). The transient change in isotherm shape shown schematically in Figure 4 was experimentally observed by D'Souza et al.³⁵ using a similar Bridgman apparatus as the heat-transfer mode changes from chill conduction to mold-wall radiation. Therefore, the influence

of these transient curved isotherms was simulated to investigate stray grain formation near the mold wall during seed melt-back, together with heterogeneous nucleation.

After the creation of a seed microstructure (Figure 5a), the temperature was increased above the liquidus with a convex isotherm imposed to partially melt the seed. The seed was held for a 10 s soak before withdrawal (Figure 5b). As withdrawal began, the isotherms smoothly transitioned from convex to concave (Figure 5c and 5d). The morphological differences between the structures above and below the convex interface (Figure 5b and 5c) are similar to those observed experimentally by Stanford et al.³¹ in DS bars quenched during melt-back. This structural change is due to the coarsening during the transition from melt-back to withdrawal, followed by the relatively rapid growth of dendrites near

the mold wall where there is a high level of undercooling due to the much more rapid cooling than near the middle after withdrawal starts.

For the simulation shown in Figure 5, the second primary from the mold wall (on the left) blocks the first primary by the rapid lateral growth of a secondary arm (Figure 5c). At an even later stage (Figure 5d), the second primary blocks the third one due to the enhanced growth of its secondaries in the concave isotherms. This increases the primary arm spacing near the mold wall, in agreement with industrial observations with larger arm spacing found in the outer region of the seed with axial orientation after melt-back.³⁶

Although it is a function of the initial random number selected for each simulation, no stray grains are formed in the region where the isotherms are in transition from convex to concave. Instead stray grains can be observed about 1 mm above the melt-back interface (Figure 5d). For this simulation, this region was not undercooled long enough to reach the required probability for a nucleus in this area to be activated. This is due in part to the fast growth of the dendrite tip and in part to a perhaps unrealistic assumption in the simulation—that there is no fluid flow. As the molten metal is poured into the mold above the seed, it might remove some of the solute in this region, increasing the effective undercooling.

The stray grains that are nucleated 1 mm to 2 mm above the melt-back interface area are overgrown by the primaries from the seed in most cases where the seed <001> direction is aligned to the axis of the casting.

Influence of the Seed's Crystallographic Orientation

As can be seen from Figure 2, the stray grains overgrow the seed primaries only on one side of the casting when off-axis seeds are used. It was therefore hypothesized that the orientation of the seed primaries relative to the mold wall (converging or diverging) may influence the competitive growth mechanism between the seed and stray grains. Therefore, seeds with axial and $\pm 15^\circ$ off-axial orientations are simulated.

The number and survival of stray grains formed was found to be a strong function of seed orientation (see Figure

Interface Curvature

The Piecewise Linear Interface Calculation (PLIC) algorithm, which was originally developed for fluid flow problems,²³⁻²⁵ was used to reconstruct the interface from the fraction solid field. The PLIC method assumes that the interface can be approximated as a straight line inside any given interfacial cell. The normal to the interface, n , is then given by

$$n = \nabla f_s / |\nabla f_s| \quad (D)$$

Equation D is calculated using finite difference gradient method of Parker and Youngs.²⁶

The curvature of the interface, k , is then defined in terms of the divergence of the unit normal, $\hat{n}$,^{27,28}

$$k = \nabla \cdot \hat{n} \quad (E)$$

where $\hat{n} = n / |n|$.

This curvature was then used to determine the additional undercooling for all interface cells as described by Wang et al.²²

Simulation Parameters

A domain size of $3 \times 6 \text{ mm}^2$ was found to be sufficient to approximate the area where stray grains were observed experimentally in the seed. This domain size was therefore used in all the simulations. An initial fixed dendrite spacing of $350 \mu\text{m}$ was used to grow the seed with orientation relative to the isotherms varied.

The equilibrium data were obtained from the linearized phase diagram for Ni-6.5 wt.% Ta. Key values used are: partition coefficient, $k = 0.70$; diffusion coefficient, $D_L = 10^{-9} \text{ m}^2\text{s}^{-1}$,²⁹ and liquidus slope, $m = -4.74$. This was considered appropriate since k in the binary system is similar to the partitioning of tantalum in CMSX4 as reported elsewhere,³⁰ and the solidification characteristics will be controlled by the heavy refractory additions having sluggish diffusion coefficients D_L .

The implementation of heterogeneous nucleation can be found in a prior publication.¹⁶ A zero-flux boundary condition was applied to the left and right sides of the domain. A withdrawal velocity as a function of time was built in to simulate the temperature profile during soak and initial withdrawal of the seed with imposed temperature gradient of 3 K/mm . The whole procedure consists of three stages: growth of the seed, partial melt-back of the seed, and withdrawal. An initial pulling velocity of $100 \mu\text{m/s}$ was used to grow a seed with desired primary arm spacing and orientation. Either flat or curved isothermal fields were imposed upon the domain to simulate the thermal conditions experienced during the melt-back process.

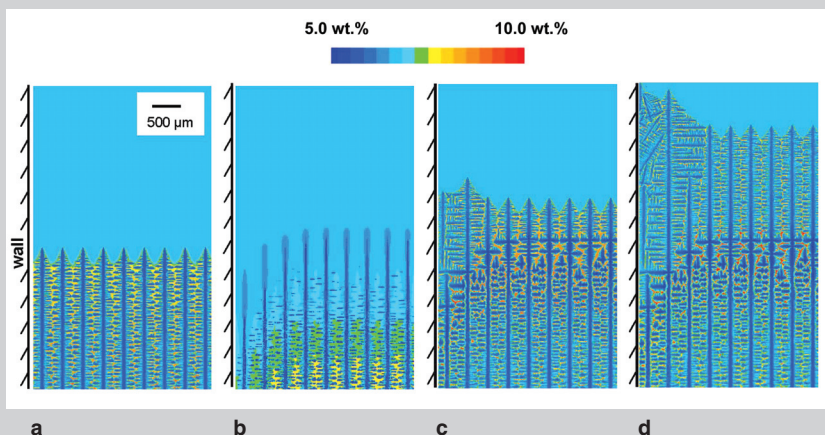


Figure 5. Predicted microstructures during seed melt-back for curved isotherms and heterogeneous nucleation: (a) seed, (b) melt-back, and (c, d) at 20 s and 45 s after withdrawal started.

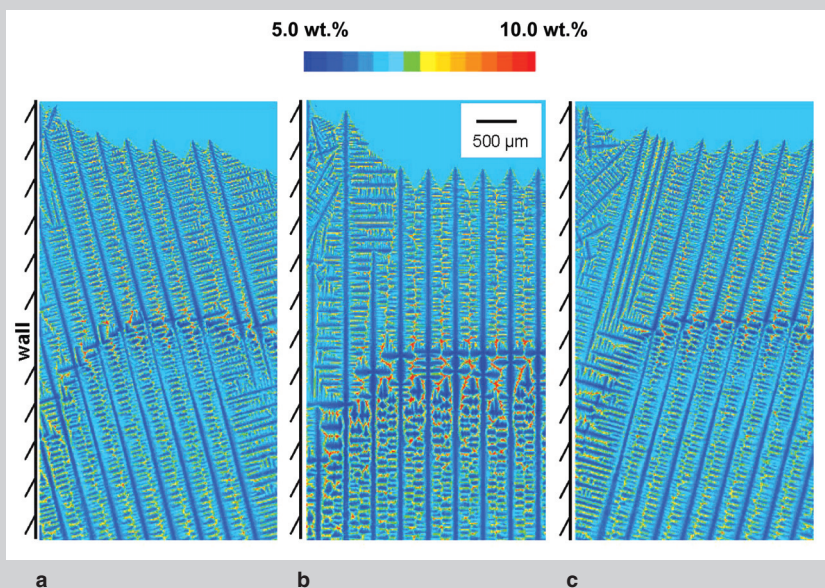


Figure 6. The predicted microstructure for different seed crystallographic orientations: (a) converging, (b) axial, and (c) diverging.

6). When the seed dendrites converge onto the mold wall only three stray grains form and all are quickly overgrown (Figure 6a). For the axial case (Figure 6b), six stray grains form and are overgrown, although more slowly. When the seed primaries diverge from the wall (Figure 6c), nine stray grains form and one continues to grow. Therefore, the survival of stray grains is more likely at the edge of the casting where the primaries diverge away from the mold wall. The stray grains also occur earlier (closer to the melt-back interface) for the diverging case as compared to when the dendrites converge onto the wall.

ACKNOWLEDGEMENTS

This study was supported in part by the Engineering and Physical Sciences Research Council (Grant No. GR/R78992), Rolls-Royce plc, and Cannon Muskegon.

References

1. K. Harris and J.B. Wahl, *Superalloys 2004*, ed. K.A. Green et al. (Warrendale, PA: TMS, 2004), pp. 45–52.
2. A.D. Cetel and D.N. Duhal, *Superalloys 1988* (Warrendale, PA: TMS, 1988), pp. 235–244.
3. E.W. Ross and K.S. O'Hara, *Superalloys 1996*, ed. E.A. Loria (Warrendale, PA: TMS, 1996), pp. 19–25.
4. S. Tin, T.M. Pollock, and W.T. King, *Superalloys 2000*, ed. T.M. Pollock et al. (Warrendale, PA: TMS, 2000), pp. 201–210.

5. N. Stanford et al., in Ref. 1, pp. 515–524.
6. K.A. Jackson et al., *Trans. Metall. Soc. AIME* (New York: AIME, 1966), vol. 236, pp. 149–157.
7. C.J. Paradies, M.E. Glicksman, and R.N. Smith, *First International Conference on Transport Phenomena in Processing*, ed. S.I. Guceri (Lancaster, PA: Technomic, 1993), pp. 266–273.
8. P.D. Lee and J.D. Hunt, *Acta Mater.*, 49 (2001), pp. 1383–1398.
9. R.C. Atwood and P.D. Lee, *Metall. Mater. Trans. B-Proc. Metall. Mater. Proc. Sci.*, 33 (2002), pp. 209–221.
10. P.D. Lee et al. *Mater. Sci. Eng. A*, 328 (2002), pp. 213–222.
11. M. McLean, P.D. Lee, and B.A. Shollock, *Advanced Materials and Processes for Gas Turbines*, ed. G.E. Fuchs et al. (Warrendale, PA: TMS, 2003), pp. 83–90.
12. X. Xu, W. Zhang, and P.D. Lee, *Metall. Mater. Trans. A*, 33 (2002), pp. 1805–1815.
13. A. Kermanpur et al., *JOM*, 56 (3) (2004), pp. 72–78.
14. W. Wang et al., *Int. J. Cast Metals Research*, 15 (2002), pp. 269–271.
15. R.C. Atwood and P.D. Lee, *Acta Mater.*, 51 (2003), pp. 5447–5466.
16. X.L. Yang et al., *Mater. Sci. Eng. A*, 386 (2004), pp. 129–139.
17. H.B. Dong et al., *J. Mater. Sci.*, 39 (2004), pp. 7175–7184.
18. H.B. Dong and P.D. Lee, *Acta Mater.*, 53 (2005), pp. 659–668.
19. P.D. Lee et al., *Mater. Sci. Eng. A*, 365 (2004), pp. 57–65.
20. X.L. Yang et al., in Ref. 1, pp. 951–958.
21. D.M. Maijer et al., *Metall. Mater. Trans. A*, 35 (2004), pp. 3275–3288.
22. W. Wang, P.D. Lee, and M. McLean, *Acta Mater.*, 51 (2003), pp. 2971–2987.
23. J.W. Rider and D.B. Kothe, *J. Comput. Phys.*, 141 (1998), pp. 112–152.
24. D. Gueyffier et al., *J. Comput. Phys.*, 152 (1999), pp. 423–456.
25. J.E. Pilliod and E.G. Puckett, *J. Comput. Phys.*, 199 (2004), pp. 456–502.
26. B.J. Parker and D.L. Youngs, *Technical Report 01/92* (Berkshire, U.K.: Atomic Weapons Establishment, Aldermaston, February 1992).
27. A. Jacot and M. Rappaz, *Acta Mater.*, 50 (2002), pp. 1909–1926.
28. H. Tang, L.C. Wrobel, and Z. Fan, *Comput. Mater. Sci.*, 29 (2004), pp. 103–118.
29. M.G. Ardakani et al., *Superalloys 2000*, ed. T.M. Pollock et al. (Warrendale, PA: TMS, 2000), pp. 219–228.
30. A. Wagner et al., *Proceedings of the 2001 International Symposium on Liquid Metal Processing and Casting*, ed. A. Mitchell (Warrendale, PA: TMS, 2001), pp. 301–313.
31. N. Stanford et al., *Scr. Mater.*, 50 (2004), pp. 159–163.
32. J.D. Hunt, *Mater. Sci. Eng. A*, 65 (1984), pp. 75–83.
33. M. Gaumann, R. Trivedi, and W. Kurz, *Mater. Sci. Eng. A*, 226 (1997), pp. 763–769.
34. C.Y. Wang and C. Beckermann, *Metall. Mater. Trans. A*, 25 (1994), pp. 1081–1093.
35. N. D'Souza et al., *Metall. Mater. Trans. A*, 31 (2000), pp. 2877–2886.
36. M. Newell, Internal communications on seed melt-back (Rolls-Royce plc, 2005).

X.L. Yang and P.D. Lee are with the Department of Materials at Imperial College London. N. D'Souza is with the Precision Casting Facility, Rolls Royce plc, in Derby, United Kingdom.

For more information, contact P.D. Lee, Department of Materials, Imperial College London, Exhibition Road, London SW7 2BP, U.K.; e-mail p.d.lee@imperial.ac.uk.